

- (i) The variation of V_{IN} with t can be represented by

$$V_{\text{IN}} = A \cos Bt$$

where A and B are constants.

Determine the values of A and B . Give a unit with your answer for A .

$$A = \dots\dots\dots \text{unit} \dots\dots\dots$$

$$B = \dots\dots\dots \text{rad s}^{-1} \quad [2]$$

- (ii) Determine the type of rectification produced by the circuit in Fig. 6.1.

..... [1]

- (iii) On Fig. 6.3, draw the circuit diagram for the components inside the rectification circuit.

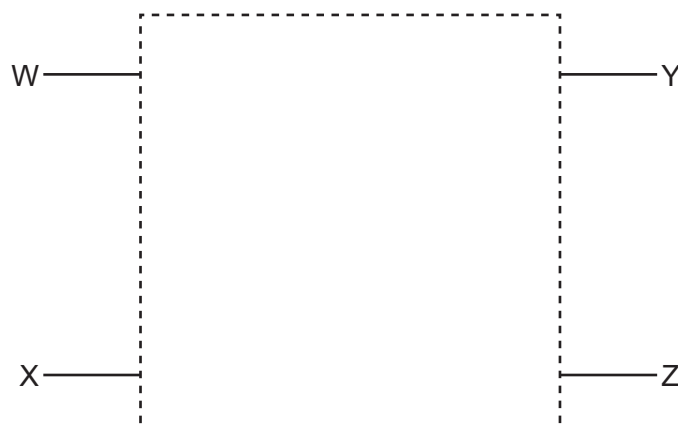


Fig. 6.3

[2]

- (iv) Determine a value for the time constant for the discharge of the capacitor C through the resistor R in Fig. 6.1.

time constant = s [3]

